

FarSightBERT: Enhanced Embeddings for Long-Term Disease Prediction

Nikhil Kapila
nkapila6@gatech.edu

Tejas Rath
trathi9@gatech.edu

[Data](#) | [Video](#) | [GitHub Repository](#)

Abstract—The FarSight paper by Gangavarapu et al [1] demonstrated the value of unstructured clinical notes for long-term disease prediction showcasing how rich patient-specific information is often lost in structured EHR data. Building upon their innovative FarSight aggregation technique, we explore two significant extensions to their approach: use of transformer-based embeddings and mixture of expert models.

Index Terms—Clinical decision support systems, Disease prediction, Healthcare analytics, ICD-9 code group prediction, Precision medicine

I. INTRODUCTION

The healthcare industry generates vast amounts of clinical data ranging from structured electronic health records (EHR) to unstructured clinical narratives [2]. Focus of predictive modeling on clinical data has been in the EHR landscape. Unstructured text especially nursing notes contains rich, patient-specific information, captures nuanced observations are ignored in predictive modeling tasks.

Gangavarapu et al.'s FarSight [1] aggregation approach is an advancement in modeling unstructured nursing notes for disease prediction. Their work demonstrated how nursing notes when properly cleaned and aggregated can predict ICD-9 diagnostic codes with remarkable accuracy. The key idea of their FarSight aggregation mechanism was the ability to associate early clinical observations with eventual diagnoses which enabled detection of diseases at their earliest manifestations.

II. RECENT WORK

Since the publication of FarSight in 2020, significant advancements have been made in leveraging unstructured clinical notes for predictive modeling tasks.

- **Representation techniques:** The space has moved from simple vector space models such as Doc2Vec and NMF to more sophisticated contextual embeddings from transformer models like BioCLinicalBERT, ClinicalBERT, and PubMedBERT.
- **Model architectures:** Deep learning architectures have evolved from basic ConvLSTM networks to sophisticated transformer models.
- **Integration of temporal dynamics:** More recent approaches explicitly model the implicit temporal dynamics in clinical notes. Models such as Temporal Graph Transformers (TRANS) [3] represent patient history as a temporal heterogeneous graph while frame-

works like TIMER [4] focus on temporal instruction tuning for longitudinal clinical records.

- **Foundation models and LLMs:** Latest developments include clinical foundation models like CLMBR [5] and adaptation of general-purpose LLMs through frameworks like CPLLM [6].

Research has expanded beyond FarSight's focus on ICD-9 code prediction to tasks such as: ICD-10 code predictions, mortality prediction, disease-specific prediction. Furthermore, research has validated FarSight's premise on value of nursing notes, with some studies finding that models trained solely on nursing notes outperform those trained on physician notes for certain tasks [7].

A. Persistent challenges

Despite significant advanced and progress of research, several challenges still exist:

- **Interpretability:** As models grow complex, providing clinically meaningful explanations become increasingly difficult yet critical for adoption.
- **Generalization:** Models trained on a single-institution data cannot be democratized and raises concerns about performance in new environments.
- **Temporal distribution shift:** Performance degradation over time due to evolving clinical practices and patient performance is an implicit given.

The field continues to evolve rapidly, with current research increasingly focusing on addressing these challenges while pushing boundaries of prediction performance.

III. SCOPE OF REPRODUCIBILITY AND NOVELTY

As artificial intelligence in the text modality has continued to evolve, new opportunities emerge to enhance these approaches. In this study, we build upon the foundation laid out by FarSight and advance use of unstructured clinical narratives for disease prediction. We investigate whether modern transformer-based language models like BioCLinicalBERT pre-trained on a vast corpora of biomedical text can offer richer semantic representations than document embedding techniques used in the original FarSight study. Furthermore, advances in neural architecture design especially mixture of expert models, present new possibilities for creating interpretable and efficient prediction systems.

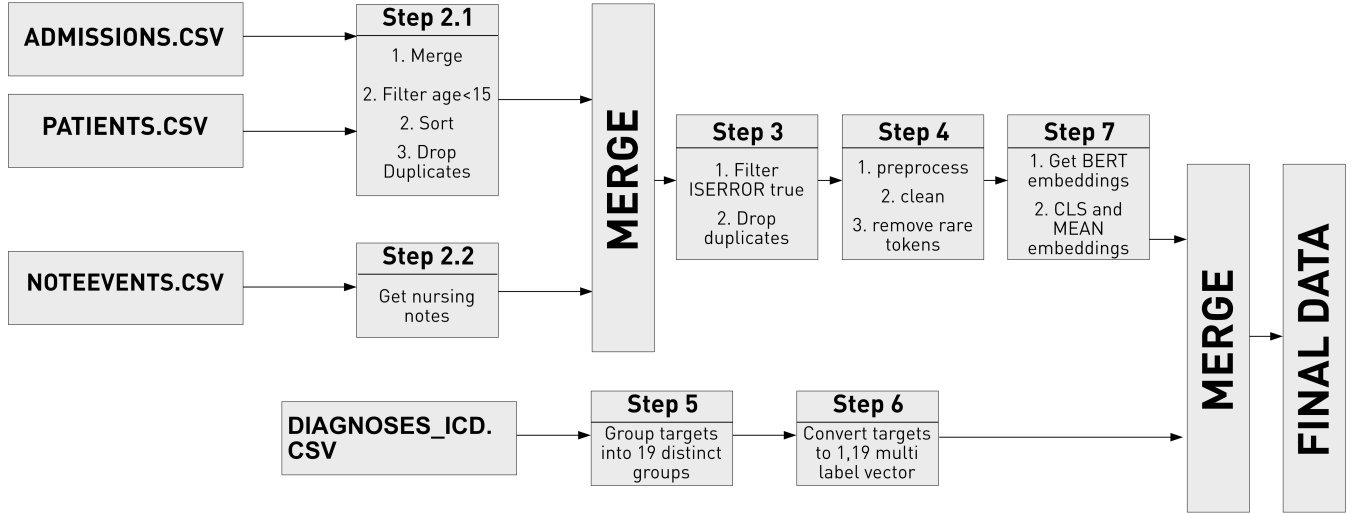


Fig. 1: Data cleaning and preprocessing procedures performed on the MIMIC-III dataset are aligned with the step numbering used in the accompanying codebase available on the GitHub repository.

Our work contributes to ongoing effort to develop more sophisticated clinical systems that can leverage full spectrum of available patient data by combining FarSight’s aggregation model with cutting-edge language models and neural architectures.

A. Transformer-Based embeddings vs Corpus Specific Representations

The original FarSight paper explored several text to vector modeling approaches for clinical nursing notes, focusing on Doc2Vec and various configurations of Non-negative Matrix Factorization. Their best results came from applying NMF to Term-Weighted (TF-IDF) matrices with Semantic Coherence (SC) using 100 topics.

These approaches derive representations directly from the text itself whereas our study investigates potential of transfer learning through pre-trained transformer encoder-only models, specifically BioClinicalBERT [8]. BioClinicalBERT is based on BioBERT [9] which is a domain-specific version of BERT [10] pretrained on biomedical literature such as PubMed abstracts and PMC full texts. BioClinicalBERT extends BioBERT by pretraining on the large unstructured notes in the MIMIC-III [11] dataset and is publicly available on HuggingFace [8].

Unlike the static embeddings from vector modeling in FarSight, transformer encoder only models can generate contextual embeddings where a word’s representation changes based on its surrounding context. This capability allows for more nuanced understanding of medical terminology.

We hypothesize that transformer-based contextual-aware embeddings from BioClinicalBERT will help overcome limitations observed in the original study’s modeling approaches. We anticipate that these embeddings would: (1) capture more nuanced semantic relationships compared to Bag-of-Words and Doc2Vec, (2) better handle complex, specialized terminology in nursing documentation and (3) leverage the knowledge acquired during pre-training on a large medical corpus.

B. Mixture of Expert (MoE) models

We further extend FarSight’s approach by investigating potential of Mixture of Expert (MoE) architectures that can enhance both performance and interpretability. The FarSight study identifies several challenges in dealing with clinical notes, including their longitudinal aspect, heterogeneity, voluminous content, and complex structure. The main idea behind using MoE is inspired to handle heterogeneity by specialization.

Each model (expert) optimizes on its own inductive biases and hence, we expect each model to specialize on different aspects of the data. For e.g., ConvLSTMs may be better at capturing temporal pattern in notes while ConvNet may be better at identifying specific keywords in shorter, acute-event notes.

The MoE learns to combine strengths of different neural architectures to make the best overall prediction. Furthermore, this modeling approach gives potential for interpretability as we can analyze which architecture the gating network relies on most for different type of notes. This can provide insight into which architectural biases can be most useful for different clinical scenarios.

IV. DATA METHODOLOGY

A. Dataset description

MIMIC-III v1.4 [12] is a publicly available, de-identified database comprising detailed health-related data from over 40,000 patients admitted to critical care units at the Beth Israel Deaconess Medical Center in Boston, Massachusetts, between 2001 and 2012. The MIMIC-III v1.4 database consists of 2,083,180 note events out of which 223,556 are nursing notes from 7,704 distinct patients. We model on these nursing notes which is a huge a corpus of 5,244,541 sentences, 79,988,065 total words and 715,821 unique words.

B. Dataset Preprocessing and Cleaning

An overview on our dataset preprocessing and cleaning can be seen in Fig. 1. As seen in Fig. 1, we only use 4 CSV files from the entire MIMIC-III dataset.

a) *Cohort selection*: Our cohort selection approach is same as the one illustrated in the FarSight paper. We do the following: (1) we filter out neonates (age < 15), (2) keep only first ICU admissions for each MIMIC-III subject and discard later admissions, (3) identify and filter out any nursing notes with clerical error attribute, and (4) remove duplicate patient records.

The authors do (1) to maintain consistency in benchmarking with respect to related works, (2) was done because the authors found that for 94% of patients, diagnostic code groups in first admissions overlapped with those in later ones. Furthermore, using only first admissions helps avoid statistical complications that arise from having multiple admissions from same patient, this also helps in decreasing computational requirements without significant loss of information.

b) *Text preprocessing*: Our text preprocessing steps differs a little bit compared to the FarSight paper. Once the cohort selection has completed, we preprocess the text data in the nursing notes. We do the following: (1) the text is split into individual tokens using NLTK tokens, (2) common stopwords are eliminated using NLTK English stopwords corpus, (3) we remove any punctuation marks, (4) text is converted to lowercase, (5) stemming and lemmatization is applied, and (6) tokens appearing in fewer than 10 nursing notes are eliminated.

We deviate from the original paper by omitting medical abbreviation disambiguation. The CARD-2 disambiguation framework used in the original implementation was not publicly available. While we are unable to directly assess the impact of this omission, we rely on BioClinicalBERT to implicitly resolve abbreviation disambiguation through its contextualized representations.

c) *FarSight Data Aggregation*: We apply FarSight’s aggregation mechanism to map each nursing note to all ICD-9 diagnostic code groups observed in that patient’s admission. This enables detection of disease onset with early symptoms before any formal diagnosis.

d) *ICD-9 code grouping*: Mapped ICD-9 diagnostic codes into 19 distinct diagnostic groups based on code ranges

defined in [13]. ICD-9 code range of 760-779 corresponding to neonates (age < 15) are not part of our cohort and excluded in this study. Furthermore all reference and supplemental V-codes are grouped into same code group to lower computational complexity to train.

e) *BERT embeddings (Clinical Feature Modeling)*: We explore two distinct strategies to generate embeddings from BioClinicalBERT: (1) utilizing [CLS] token representation and implementing mean pooling across all token embeddings.

We extracted only the final hidden state of the [CLS] token as a 768-dimensional vector representing the entire nursing notes. The [CLS] token is specifically trained during BERT’s pretraining to capture sentence-level semantics, i.e. the CLS token gives a summary representation of the entire sequence and is often used for classification tasks in many modeling approaches.

However, given the unstructured and highly variable nature of nursing notes which often contain patient-specific information, we perform a mean pooling across all token embeddings that may offer a more robust representation. This method averages contextual information across the entire sequence, potentially capturing a broader semantic understanding than a single token embedding.

Therefore, we generate two sets of data: one based on [CLS] tokens and the other with mean pooled tokens to perform our downstream task.

V. MODEL METHODOLOGY

The referred FarSight paper [1] discusses various models viz. MLP, ConvNet, LSTM, Bi-LSTM, Conv-LSTM, Seg-GRU and evaluate their performance on unstructured clinical nursing notes. We choose the top 3 performing models from original paper i.e. Conv-LSTM, Bi-LSTM and ConvNet and simple MLP architecture. Conv-LSTM have consistently highest performance as seen for multiple metrics and on various type of embeddings viz. Doc2Vec, NMF-BoW, NMF-TW etc., followed by ConvNet and Bi-LSTM. All these architecture models are discussed below.

A. Bi-LSTM

LSTM (Long Short-Term Memory) is a type of Recurrent Neural Network (RNN) designed to address the vanishing gradient problem commonly observed in traditional RNNs.

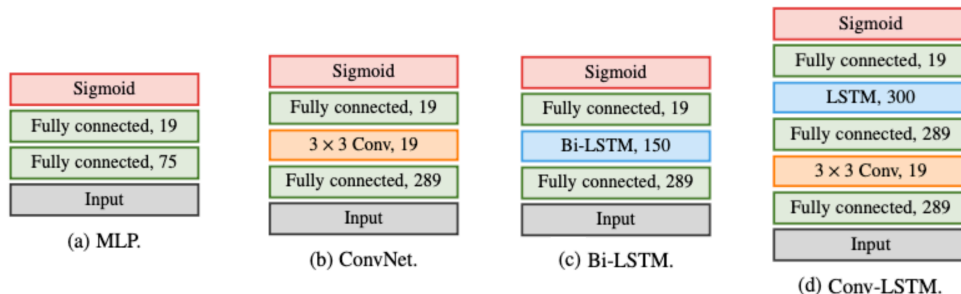


Fig. 2: Deep neural architectures used in the study. Taken from the FarSight [1] paper.

The LSTM architecture includes a gating mechanism that regulates the flow of information through the network. These gates include:

- **Forget Gate (f):** Decides which information to discard from the previous cell state.
- **Input Gate (i):** Determines which new information to update in the cell state.
- **Cell State (c):** Maintains the memory of the network, updated by the forget and input gates.
- **Output Gate (o):** Controls the output of the current cell state.

The LSTM model processes the input sequence step-by-step, maintaining a current cell state c_t and a hidden state h_t at each time step. These states are influenced by the previous cell state c_{t-1} and hidden state h_{t-1} , allowing the model to retain semantic meaning over long sequences. This capability is particularly important for clinical notes, where the meaning of terms is often influenced by preceding terms. However, the LSTM outputs only considers the past inputs for generating output.

Bi-LSTM (Bidirectional Long Short-Term Memory) extends the capabilities of LSTM by processing the input sequence in both forward and backward directions. This allows the model to capture context from both past and future terms, which is particularly beneficial for understanding the semantic meaning of terms in nursing notes.

The Bi-LSTM architecture consists of two LSTM layers:

- **Forward LSTM:** Processes the input sequence from the beginning to the end, capturing past dependencies.
- **Backward LSTM:** Processes the input sequence in reverse, capturing future dependencies.

The outputs from both the forward and backward LSTMs are concatenated at each time step to form a comprehensive representation of the input sequence.

In the context of clinical nursing notes, where the semantic meaning of terms is influenced by their surrounding context, Bi-LSTM provides a significant advantage over unidirectional LSTMs. By leveraging information from both directions, Bi-LSTM improves the model's ability to make accurate predictions based on the input data.

In our implementation, we have fixed the architecture as shared in [1]. A schematic overview of the architecture is presented in the accompanying image. The Bi-LSTM model is configured with the following specifications:

- **Number of LSTM layers:** 1
- **Linear Layer:** A fully connected layer reduces the 768 dimensional BioBERT embeddings to a latent space of 289 dimensions. This layer acts like a dimensional reduction step while aiming to retain semantic richness, i.e. extract most information from 768 input dimension.
- **Hidden State Size:** 300 units per direction, resulting in a total of 600 when combined due to bidirectionality.
- **Bi-LSTM Module:** A single-layer bidirectional LSTM processes the reduced embeddings to capture both forward and backward contextual dependencies.
- **Embedding Layer:** A fully connected layer is placed before the Bi-LSTM module to reduce the dimensionality of BERT's 768 dimensional contextualized

embeddings to 289 while retaining maximum semantic information.

- **Output Layer:** A final fully connected layer maps the 600-dimensional BiLSTM to 19 output classes.

B. ConvNet, Convolutional Neural Network

CNN has proven to be an efficient architecture to process image data. CNN utilizes convolving filters (kernels) to extract meaningful features from input data. We extend these capabilities of CNN to textual modality. Each filter is responsible for extracting one feature; multiple filters can be combined to fetch multiple features.

Under the hood, the input is a 768-dimensional vector derived from BioClinicalBERT embeddings. A convolution operation involving a filter is applied to a window of h terms to produce a new feature. These features are applied to every possible window of terms in embeddings.

The ConvNet architecture used in this study is configured as follows:

- **Input Layer:** A fully connected layer reduces the 768-dimensional BioBERT embeddings to a latent space of 289 dimensions. This layer acts like a dimensional reduction step while aiming to retain maximum semantic information.
- **Reshaping step:** The 289 dimensional vector is reshaped into a 2D grid of shape (1, 17, 17) to simulate a spatial structure for 2D convolution. This is purely a data wrangling step for 2D convolution and does not reflect any real spatial locality.
- **Convolutional Layer:** A 2D convolution layer applies 19 filters of size 3×3 with stride 1 producing 19 feature maps.
- **Activation Function:** ReLU is applied after the convolutional layer to introduce non-linearity and promote sparse activation.
- **Output Layer:** Output feature maps are flattened and passed through a fully connected layer to produce the logits to output 19 target classes.

This architecture is particularly effective for capturing local dependencies in textual data, making it well-suited for analyzing clinical nursing notes. The ConvNet model complements the Bi-LSTM by focusing on local patterns, while the Bi-LSTM captures long-term dependencies.

C. Conv-LSTM

As described in ConvNet, convolutional layer extracts high level features from given BERT embeddings of clinical nursing notes. But it cannot capture long term dependencies in nursing notes. The idea for this architecture is to capture the capabilities of Convolution and LSTM. The hybrid architecture is efficient in capturing high level features as well as retains long term dependencies.

The ConvLSTM architecture used in this study is configured as follows:

- **Input Layer:** A fully connected layer reduces the 768 dimensional BioBERT embeddings to a latent space of 289 dimensions. This layer acts like a dimensional re-

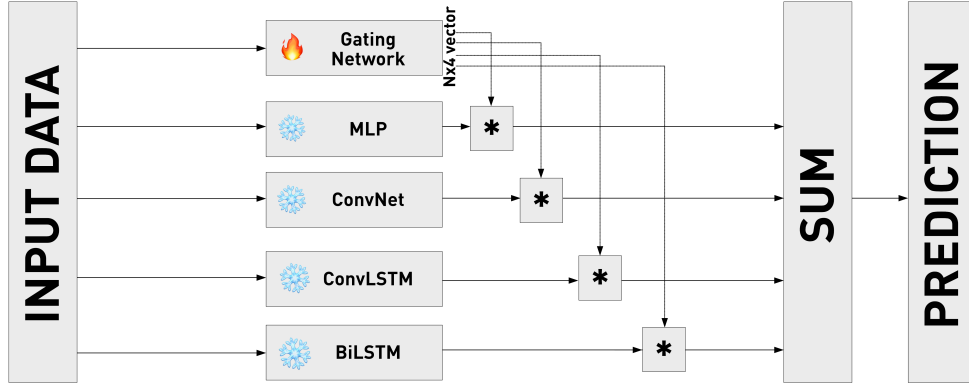


Fig. 3: A heterogeneous MoE model architecture. Input data is processed in parallel by multiple expert networks. The gating network assigns dynamic weights to each expert’s output which are then combined via weighted summation to produce the final prediction. Pretrained models are frozen, and only gating network is fine-tuned to learn which expert contributes most to the output.

duction step while aiming to retain maximum semantic information.

- **Reshaping step:** The 289 dimensional vector is reshaped into a 2D grid of shape (1, 17, 17) to simulate a spatial structure for 2D convolution. This is purely a data wrangling step for 2D convolution and does not reflect any real spatial locality.
- **Convolutional Layer:** A 2D convolution layer applies 19 filters of size 3×3 with stride 1 producing 19 feature maps.
- **Intermediate Linear Layer:** Accepts 19 flattened feature maps and transforms those to 289 dimensional latent space.
- **LSTM Module:** The resulting 289 dimensional vector is reshaped to a 3D tensor and passed through a single layer LSTM with a hidden state size of 300. Although the input is non-sequential, the LSTM acts as a non-linear aggregator enabling richer interactions across transformed feature space.
- **Output Layer:** Output from LSTM module is passed through a fully connected layer to produce the produce the logits to output 19 target classes.

D. Mixture of Experts (MoE) model

As seen in Section III.B, we employ a Mixture of Experts (MoE) approach to effectively handle the heterogeneity present in clinical nursing notes. Rather than treating all notes identically, we leverage multiple specialized models trained on BioClinicalBERT embeddings, each potentially capturing distinct aspects of the clinical narrative.

The conceptual foundation of our approach draws from the seminal work of [14], which established that conditionally routing inputs to specialized expert models can improve performance on heterogeneous data. We implement this through a gating network as illustrated in Fig. 3, which dynamically determines the optimal weighting of each expert model based on the characteristics of each incoming clinical note.

This adaptive ensemble strategy allows our system to route different types of nursing documentation to the most appropriate expert models. For instance, notes containing

primarily vital sign observations might be directed toward one expert, while notes focused on medication responses might be weighted toward another. The gating network learns these routing patterns automatically during training, effectively discovering latent note types and their corresponding optimal prediction strategies without requiring manual categorization.

VI. TRAINING

A. Computational implementation

All code is written in a Python 3.10 environment. Data preprocessing and cleaning pipeline was done locally using Pandas, training and modeling architecture were written in PyTorch 2.6.0 and the sentence-transformers library was used to extract BERT embeddings from BioClinicalBERT.

Our experiments were conducted on an Apple M1 Pro and CPU hardware configurations. Despite using consumer grade computing resources, we achieved reasonable training efficiency with an average runtime of approximately 50-60 seconds per epoch resulting in a total training time of around 6.5 minutes per model.

The extraction of FP32 embeddings for both CLS token and mean pooling methods took approximately 1.5 hours each on an Apple M1 Pro.

B. Training details

- **Dataset:** Two datasets are generated as seen in Section IV.B.e. We use BioClinicalBERT to generate datasets based on [CLS] tokens and mean pooled tokens.
- **Train-Test Set:** The preprocessed dataset is split into training and testing sets. 80% of the records are allocated to the training set, while the remaining 20% are used for testing. This split ensures that the model is trained on a majority of the data while retaining a portion for evaluating its performance.
- **Hyperparameter Tuning:** We initially implemented a multi-objective Bayesian optimization approach that focused on increasing validation accuracy and decreasing the generalization gap. However, given that

such optimization requires numerous trials and our primary focus was on comparing embedding methodologies rather than hyperparameter optimization, we maintained fixed hyperparameters across experiments. Following the original paper, we used a batch size of 128, 8 epochs and a learning rate of $1e-3$.

- **Optimizer:** Adam optimizer is used to train the model. Adam (Adaptive Moment Estimation) automatically adjusts the learning rate for each parameter based on estimates of the first (mean) and second (variance) moments of the gradients. This enables faster and more stable convergence, especially in high-dimensional spaces. Compared to traditional optimizers like SGD, Adam is robust, more computationally efficient, and generally converges faster on large and complex datasets.
- **Random State:** To ensure the reproducibility of experiments, random state is set for all the training and experiments.
- **Loss Function:** The task of ICD-9 code prediction is multi-label classification problem, i.e. each clinical record can have multiple labels. In this setting, each label is treated as an independent binary classification task. To accommodate this, the model uses Binary Cross Entropy with Logits Loss (BCEWithLogitsLoss) which computes the binary cross-entropy loss independently for each class and averages the result across all classes and samples. The loss is calculated as shown in Equation (1).

$$\begin{aligned} \text{BCEWithLogitsLoss} = & \\ -\frac{1}{N * C} \sum_{i=1}^N \sum_{c=1}^C & \left[w_c * (y_{\{i,c\}} * \log(\sigma(z_{\{i,c\}}))) \right. \\ & \left. + (1 - y_{\{i,c\}}) * \log(1 - \sigma(z_{\{i,c\}})) \right] \end{aligned} \quad (1)$$

Variables:

- N : Number of samples in the batch.
- C : Number of classes.
- $z_{i,c}$: Raw logit (unnormalized output) for class c in sample i .
- $y_{\{i,c\}} \in \{0, 1\}$: Ground truth label for class c in sample i .
- $\sigma(z_{\{i,c\}})$: Sigmoid function applied to logit $z_{\{i,c\}}$.
- w_c : Optional weight for class c (e.g., pos_weight to handle class imbalance).

C. Mixture of Experts (MoE) training

Our training procedure follows a two-phase approach. In the first phase (as seen in Section VI.B), we independently train four expert models on two different BioClinicalBERT embeddings variants. Once these expert models have been trained, we freeze their parameters to preserve their specialized knowledge. In the second phase, the frozen expert models are integrated with a trainable gating network as seen in Fig. 3. The gating network consists of a shallow neural network that takes same BioClinicalBERT embeddings as input and outputs a distribution of weights across the expert models.

During this second phase of training, only parameters of gating network are updated while expert models are fixed. The final prediction is a weighted sum of the expert outputs with weights determined by the gating network. We continue to use binary cross-entropy loss with logits, maintaining consistency with our individual model training. This approach allows the gating network to learn which expert(s) to trust for different types of clinical notes, effectively creating a dynamic ensemble that outperforms any individual model.

VII. EVALUATION METRICS

It is crucial to select appropriate metric to access the performance of model. The choice of metric depends heavily on the nature of problem and characteristics of dataset (e.g., balanced vs imbalanced). Since our ICD-9 dataset is imbalanced in nature, we followed the approach in the FarSight paper and selected various standard classification evaluation metrics suited for multi-label and imbalanced datasets.

- **Accuracy:** Measures the proportion of correctly predicted labels out of the total labels. However, for imbalanced datasets, accuracy can be misleading as it does not account for the distribution of classes.
- **Matthews Correlation Coefficient (MCC):** A balanced measure that takes into account true and false positives and negatives. It is especially useful for imbalanced datasets as it provides a single score that evaluates the quality of binary classifications. The MCC value ranges from -1 to $+1$,
 - $+1$ indicates a perfect prediction
 - 0 indicates no better than random prediction
 - -1 indicates total disagreement between prediction and observation.
- **F1 Score:** Harmonic mean of precision and recall. It is particularly useful for imbalanced datasets as it balances the trade-off between precision and recall. The F1 score ranges from 0 to 1 , where 1 indicates perfect precision and recall. In our multilabel settings for this problem, weighted F1 is used to balance the impact of rare and frequent code groups.
- **Area Under the Precision-Recall Curve (AUPRC):** Evaluates the trade-off between precision and recall across different thresholds. It is a more informative measure than AUROC for imbalanced as it emphasizes on model's ability to identify true positives without being diverted by majority of true negatives.
- **Area Under the Receiver Operating Characteristic Curve (AUROC):** Represents model's ability to discriminate between different classes by plotting true positive rate against false positive rate. This metric provides measure of ranking positive instances over negative ones by the classifier. A higher AUROC score (closer to 1) indicates stronger class separability.

All these above metrics evaluate the model from a different perspective, hence ensuring that model is accurate, as well as robust, reliable and affective across all diagnostic code groups - including minor classes.

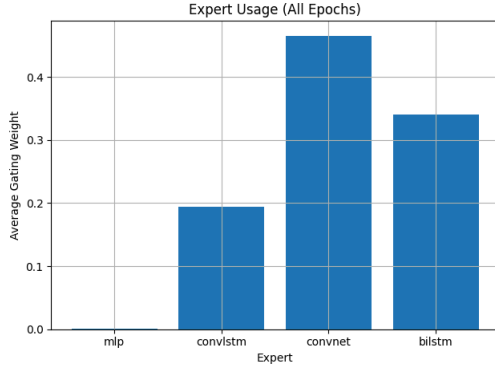


Fig. 5: Expert usage across epochs w/ models trained on CLS embeddings

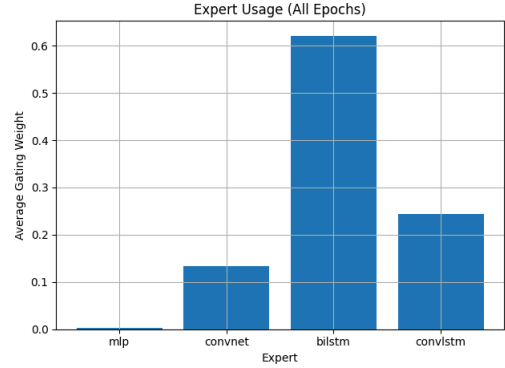


Fig. 6: Expert usage across epochs w/ models trained on Mean embeddings

VIII. RESULTS

It is interesting to see that both [CLS] token and mean pooled embeddings from BioClinicalBERT show lower performance compared to static embeddings of NMF-TW with SC from FarSight’s paper. Furthermore, CLS token slightly underperforms mean pooling on most metrics and mean pooling shows modest improvements in MCC scores and AUROC

Below are some important inferences:

- **Domain-specific representation matters:** Referring Table I, it is inevitable that domain-adapted NMF-TW with SC outperforms the pre trained BioClinicalBERT embeddings, even though BioClinicalBERT is specifically pre-trained on all MIMIC notes. This suggests that NMF-TW with SC model captured rich information in the informally-written nursing notes needed for classifier to learn and generalize; better compared to BioClinicalBERT [8].
- **FarSight aggregation provides crucial advantage:** The results Table I confirms the hypothesis and core promise mentioned in [1] that Farsight aggregation mechanism substantially provides capability to model to detect the diseases at on-set from the unstructured clinical notes.
- **MCC divergence:** The most dramatic difference observed in original paper [1] and Table I is, huge divergence in MCC scores i.e. 64.59% for best NMF-TW model vs 29.12% for best BioClinicalBERT model. This indicates the NMF-TW approach is particularly strong at handling the class imbalance in the ICD-9 dataset.
- **Mean v/s CLS embeddings:** For BioClinicalBERT, it is observed that models trained on Mean embeddings consistently outperforms models trained on Mean Embeddings. This indicates that averaging all token provides more comprehensive and complete clinical information then relying on CLS token alone.
- **The transfer learning trade-off:** A compute vs accuracy trade-off arises from our result. While contextual embeddings do not beat static embeddings by marginal percentages, it is important to take note of the compute requirements as well. While NMF-TW with SC significantly outperforms BioClinicalBERT embeddings, the computational considerations are noteworthy. NMF-

TW requires substantial upfront computation to generate topic models on the specific corpus, but enables efficient subsequent neural network training. In contrast, BioClinicalBERT leverages pre-trained weights but demands significant compute resources to generate contextual embeddings for each nursing note. This trade-off is particularly relevant in resource-constrained clinical settings. The superior performance of NMF-TW (and others) may justify its computational investment, but scenarios requiring rapid deployment might benefit from transfer learning despite lower accuracy. The optimal approach ultimately depends on accuracy requirements, available computational resources, and deployment constraints.

- **MoE outperforms individual models:** It is observed that Mixture of Experts model consistently outperforms individual neural architectures when trained on both CLS token and Mean pooling embeddings. This suggests that different experts are successfully specializing in different aspects of the prediction task. The expert usage plots Fig. 5 and Fig. 6 show clear specialization patterns, with distinct experts being consistently activated for different input patterns throughout training.
 - For MoE on Mean embeddings, BiSLTM is specialist for $\sim 60\%$ of ICD-9 classification task followed by ConvLSTM and ConvNet. Mean embeddings provide better representation of clinical information and BiLSTM considers both past and future dependencies, thus combining the best of both.
 - For MoE on CLS embeddings, ConvNet is specialist for $\sim 50\%$ of ICD-9 classification task followed by BiLSTM and ConvLSTM. CLS embeddings provide summarized representation of clinical note and ConvNet’s ability to capture local patterns proves most effective.

The results demonstrate that despite the general success of transformer models like BERT in NLP tasks, specialized approaches like NMF topic modeling with the FarSight aggregation mechanism can significantly outperform them for specific clinical prediction tasks, particularly when dealing with imbalanced data.

TABLE I: PERFORMANCE METRICS ACROSS CLASSIFIERS

Data Model	Classifier	ACC	MCC	F1	AUPRC	AUROC
NMF-TW with SC (from the paper with FarSight Aggregation mechanism)	MLP	79.61%	57.53%	71.75%	67.03%	78.56%
	ConvNet	81.92%	61.99%	74.66%	69.83%	80.77%
	ConvLSTM	83.43%	64.59%	76.02%	71.70%	81.92%
	BiLSTM	80.96%	59.98%	73.18%	68.60%	79.61%
BioClinicalBERT [CLS] Token	MLP	77.30%	20.62%	70.01%	59.66%	71.38%
	ConvNet	78.20%	25.00%	71.43%	62.45%	74.24%
	ConvLSTM	77.82%	23.72%	71.75%	61.49%	73.54%
	BiLSTM	77.95%	25.07%	71.65%	62.07%	73.93%
	MoE	78.43%	25.38%	71.89%	63.24%	75.21%
BioClinicalBERT Mean Pooling	MLP	77.99%	23.21%	71.10%	61.50%	73.66%
	ConvNet	78.42%	25.76%	71.78%	63.61%	75.39%
	ConvLSTM	78.66%	25.93%	72.05%	63.65%	75.77%
	BiLSTM	79.14%	29.09%	72.27%	65.08%	76.75%
	MoE	79.3%	29.12%	72.44%	65.67%	77.33%

IX. DISCUSSION

The reproduction of FarSight paper’s yielded valuable insights into both strengths of original approach and limitations of newer embedding techniques.

A. Reproducibility Assessment

While we had no access to source code, it was possible to reproduce FarSight’s aggregation mechanism and extend it to our niche use-cases.

a) *What Was Easy*: The FarSight aggregation mechanism was easy to reproduce and implementing it required minimal interpretation, the pre-processing steps were well-documented, neural architecture to implement was clear, evaluation metrics were fairly standard and well-defined.

B. What Was Difficult

There were a few things that were difficult to comprehend. The paper did not specify a exact threshold to convert probabilities into binary predictions. Hence, we just used 0.5 which could have swayed our results. The detailed hyperparameter settings could have helped reduce significant

tuning times and made benchmarking much easier. In the pre-processing steps, it was not possible to do abbreviation disambiguation as we could not find the framework used for it which could have again swayed our results.

X. AUTHORS CONTRIBUTIONS

Name	Contributions
Nikhil Kapila	Created full data pipeline and datasets, created two different types of BERT embeddings, bug fixes, contributed with report.
Tejas Rath	Wrote model architectures, wrote training pipelines, bug fixes, contributed with report.

XI. APPENDIX

We mainly used LLMs in the initial phase to get a high-level understanding of the paper. For coding and report-writing, we mainly constricted our LLM usage to boilerplate code (such as understanding Pandas or PyTorch training loops), and Typst document formatting.

Prompt	Description	Validation
Please analyze this paper and give it's key contributions and points.	Used mainly during literature review and exploration phase.	The LLM was mostly correct but there were cases when it hallucinated.
How to optimize .apply methods on a pandas dataframe?	Used during data preprocessing pipeline	The LLM was very accurate in its response.
How to use sentence-transformers and how does it compare to HuggingFace lib?	Used during data preprocessing pipeline	The LLM was very accurate in its response.
Write a PyTorch training loop.	Used during MoE training pipeline.	The LLM was very accurate in its response.
How to make a table span 2 columns in Typst?	Report writing.	Major hallucination unless context is given.

REFERENCES

- [1] T. Gangavarapu, G. S. Krishnan, S. S. Kamath, and J. Jeganathan, "FarSight: Long-Term Disease Prediction Using Unstructured Clinical Nursing Notes," *IEEE Transactions on Emerging Topics in Computing*, vol. 10, no. 10, pp. 1–16, 2020, doi: 10.1109/TETC.2020.2975251.
- [2] "4." Oct. 2019.
- [3] J. Chen, C. Yin, Y. Wang, and P. Zhang, "Predictive Modeling with Temporal Graphical Representation on Electronic Health Records," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, Aug. 2024, pp. 5763–5771. doi: 10.24963/ijcai.2024/637.
- [4] H. Cui *et al.*, "TIMER: Temporal Instruction Modeling and Evaluation for Longitudinal Clinical Records." [Online]. Available: <https://arxiv.org/abs/2503.04176>
- [5] E. Steinberg, K. Jung, J. A. Fries, C. K. Corbin, S. R. Pfohl, and N. H. Shah, "Language models are an effective representation learning technique for electronic health record data," *Journal of Biomedical Informatics*, vol. 113, p. 103637, 2021, doi: <https://doi.org/10.1016/j.jbi.2020.103637>.
- [6] B. Shoham and N. Rappoport, "CPLLM: Clinical Prediction with Large Language Models," *PLOS Digital Health*, vol. 3, no. 12, p. e680, Dec. 2024, doi: 10.1371/journal.pdig.0000680.

- [7] G. S. Krishnan and S. S. Kamath, "Hybrid Text Feature Modeling for Disease Group Prediction Using Unstructured Physician Notes," in *Computational Science – ICCS 2020*, V. V. Krzhizhanovskaya, G. Závodszy, M. H. Lees, J. J. Dongarra, and P. M. A. Sloot, Eds., in *Lecture Notes in Computer Science*, vol. 12140. Springer, May 2020, pp. 321–333. doi: 10.1007/978-3-030-50423-6_24.
- [8] E. Alsentzer, "Bio_ClinicalBERT." [Online]. Available: https://huggingface.co/emilyalsentzer/Bio_ClinicalBERT
- [9] E. Alsentzer *et al.*, "Publicly Available Clinical BERT Embeddings," in *Proceedings of the 2nd Clinical Natural Language Processing Workshop*, Minneapolis, Minnesota, USA: Association for Computational Linguistics, 2019, pp. 72–78. doi: 10.18653/v1/W19-1909.
- [10] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Association for Computational Linguistics, 2019, pp. 4171–4186. doi: 10.18653/v1/N19-1423.
- [11] A. E. Johnson *et al.*, "MIMIC-III, a freely accessible critical care database," *Scientific Data*, vol. 3, p. 160035, 2016, doi: 10.1038/sdata.2016.35.
- [12] A. E. Johnson *et al.*, "MIMIC-III, a freely accessible critical care database," *Scientific Data*, vol. 3, no. 1, p. 160035, May 2016, doi: 10.1038/sdata.2016.35.
- [13] TDRDATA, "Search for ICD9 Codes and Descriptions." [Online]. Available: https://web.archive.org/web/20160308161055/http://tdrdata.com/ipd/ipd_SearchForICD9CodesAndDescriptions.aspx
- [14] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, "Adaptive Mixtures of Local Experts," *Neural Computation*, vol. 3, no. 1, pp. 79–87, 1991, doi: 10.1162/neco.1991.3.1.79.